2.5D U-Net for Multiple Sclerosis Lesion Segmentation

Automatic segmentation of MS lesions from brain FLAIR MRI scans using the MSLesSeg dataset

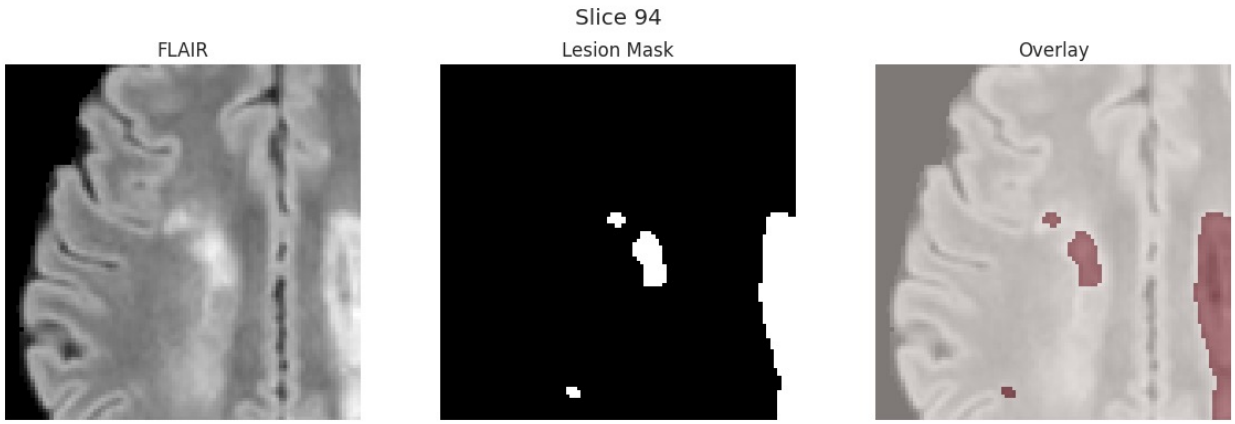
Deep Learning • Medical Image Segmentation • PyTorch

February 2026, Drury Thomas Alfio

Understanding Multiple Sclerosis and the Need for Automated Segmentation

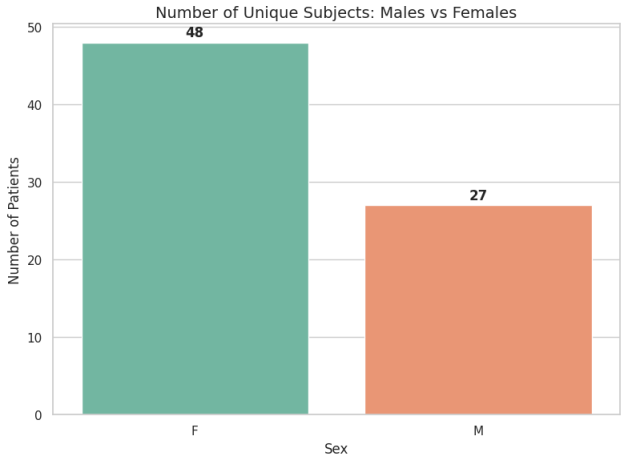
What is Multiple Sclerosis?

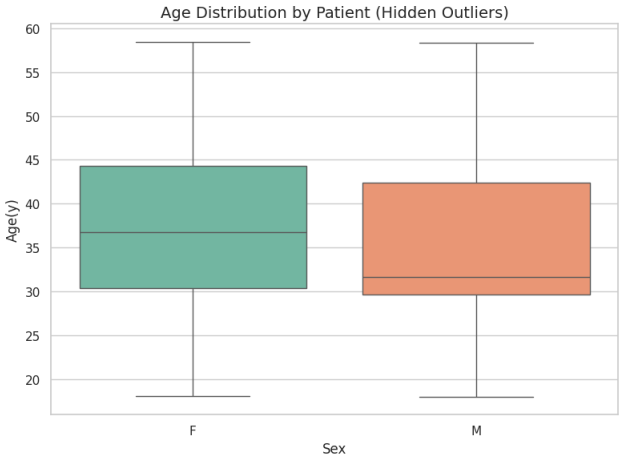
A chronic inflammatory disorder causing demyelination of the central nervous system. **FLAIR MRI** is the gold standard for visualization, where lesions appear as bright hyperintense regions.

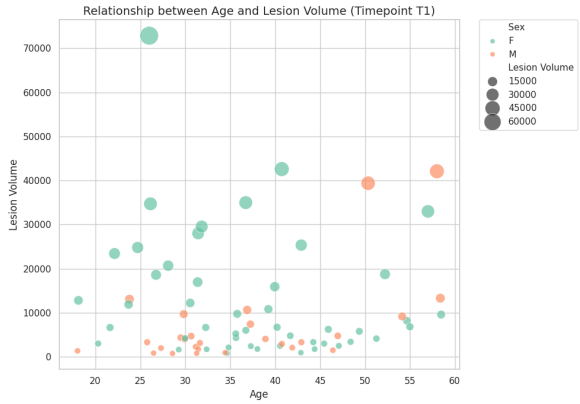


Understanding Multiple Sclerosis and the Need for Automated Segmentation

Patients overview:







2.5D Architecture Balances Context and Computational Efficiency

2D

One slice at a time

- ➕ Very lightweight & fast training
- ➖ No inter-slice context

3D

Full volume

- ➕ Best spatial context
- ➖ Extremely memory-hungry

Chosen Approach

2.5D

Stack adjacent slices

- ➕ Good context + efficient
- ➕ Fits standard GPUs
- ➖ Slightly less context than 3D

Solution: 2.5D Slice Stacking

We stack 3 adjacent axial FLAIR slices as input channels (similar to RGB). This captures inter-slice context using standard 2D convolutions.

3
Slices Stacked

15GB
GPU Memory

Model Development: Attempts

Attempt 1

2D U-Net + Threshold Filter

Dice: 0.75

- ✓ Lightweight 2D architecture
- ✓ High reported Dice score
- ✗ Test set filtered (data leakage)
- ✗ No inter-slice context

Attempt 2

2.5D U-Net, Slice-Level Split

Dice: 0.65

- ✓ 2.5D inter-slice context
- ✓ No test set filtering
- ✗ Slice-level split (leakage risk)
- ✗ No data augmentation

Final Approach

Attempt 3

2.5D U-Net, Patient Split + Aug

Dice: 0.67

- ✓ Patient-level splitting
- ✓ Data augmentation
- ✓ 2.5D context from neighbours
- ✓ No artificial test filtering

MSLesSeg Dataset Provides Robust Training Foundation

115	75	93 / 22	8,310
FLAIR MRI Scans	**Unique Patients**	**Train / Test Split**	**Training Samples**
Total Volume	Diverse Population	Strict Patient Separation	2.5D Slices

Data Preparation Strategy

No Minimum Volume Filtering

All lesions included regardless of size to ensure model robustness against small anomalies.

Strict Patient Separation

Train/test sets are completely separate patient groups to prevent data leakage.

Standardized Input

Each slice resized to 256×256 with FLAIR intensities normalized to [0, 1] range.

Data augmentation

Data augmentation is applied to train set

2.5D Slice Stacking Captures Spatial Context Efficiently

Input Stack

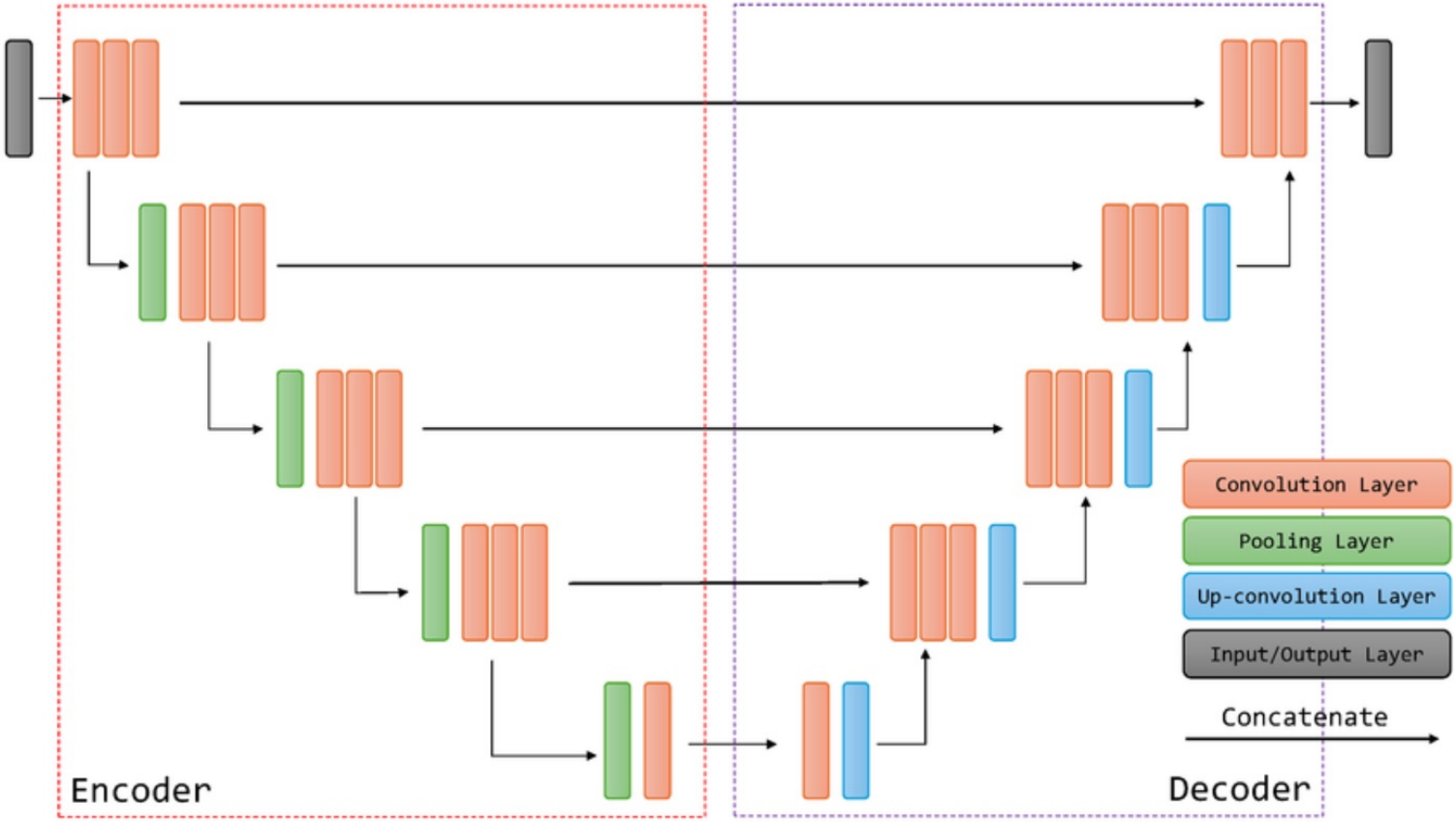
(Batch, 3, 256, 256)

3 adjacent FLAIR slices stacked as channels (similar to RGB)

Output Prediction

(Batch, 1, 256, 256)

Binary lesion probability mask for the **center slice only**

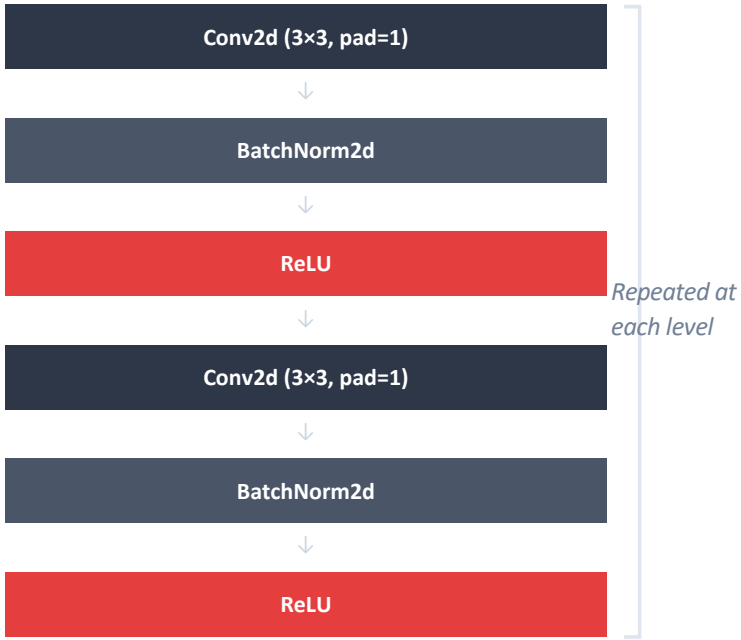


https://www.researchgate.net/figure/Illustration-of-the-U-Net-architecture-The-displayed-U-Net-is-an-encoder-decoder-network_fig1_381329058

U-net general architecture

Lightweight Design Fits Resource Constraints

DoubleConv Block Structure



Model Specifications

Parameter	Value
Total Parameters	**7,765,985**
Model Size	**~31.1 MB (float32)**
Base Features	**32**
Encoder Levels	**4**
Input Channels	**3 (adjacent slices)**
Resolution	**256 × 256**

Key Design Decisions

- ✓ Intentionally lightweight to fit Colab T4 free tier
- ✓ Batch size 16 achieves good GPU utilization
- ✓ Skip connections preserve fine spatial details
- ✓ BatchNorm stabilizes training and improves convergence

Training Configuration Optimizes Learning Dynamics

Loss Function

Binary Cross-Entropy

Optimizer

Adam

Learning Rate = 1e-4

Epochs

20

No early stopping

Batch Size

16

Validation Split

80 / 20

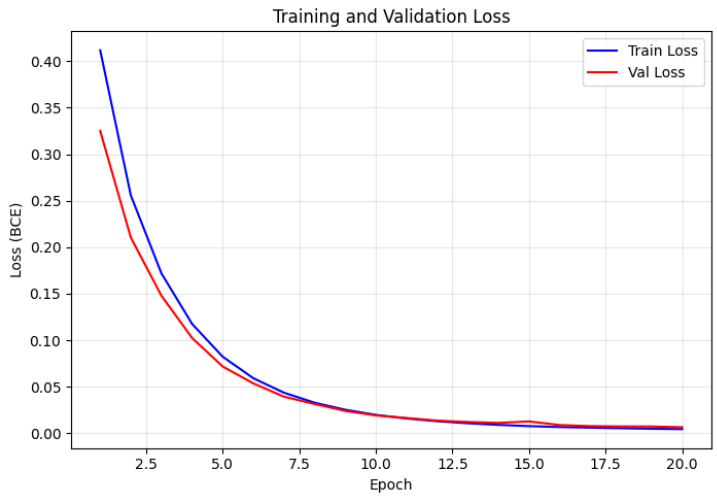
Train / Validation

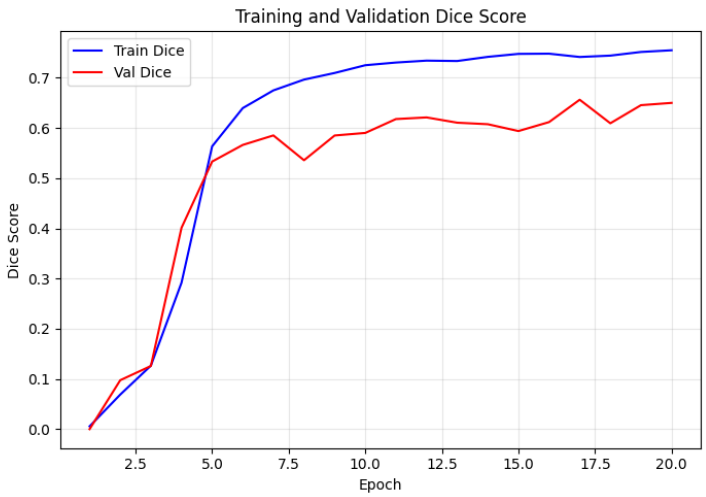
Hardware

Tesla T4 GPU

15.6 GB VRAM

Validation Performance





Epoch	Train Loss	Train Dice	Val Loss	Val Dice
1	0.4117	0.0058	0.3251	0.0002
5	0.0823	0.5634	0.0718	0.5331
10	0.0197	0.7249	0.0191	0.5901
15	0.5901	0.7475	0.0127	0.5938
17	0.0060	0.7413	0.0076	**0.6562**
20	0.0044	0.7547	0.0065	0.6499

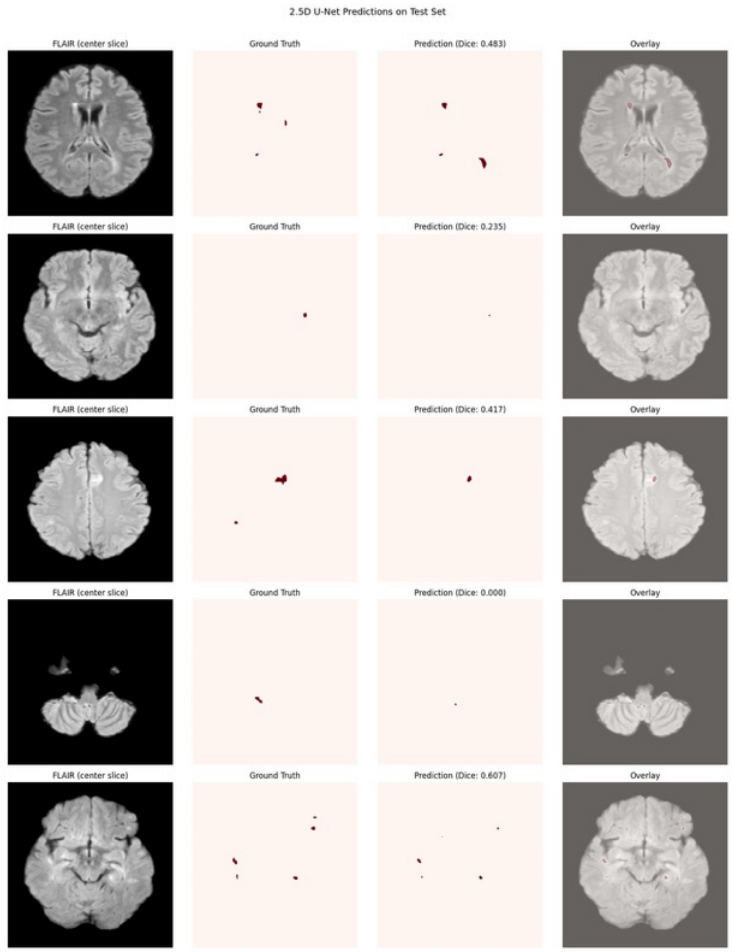
Test Results Reveal Generalization Challenges

0.6657
Dice Score

0.6982
Precision

0.5082
IoU

0.6725
Sensitivity/Recall



Analysis Identifies Strengths and Improvement Opportunities

Model Strengths

- **Strong Validation Performance:** Achieved 0.66 Dice score during training phase.
- **High Peak Accuracy:** Several test patients showed excellent segmentation (>0.75 Dice).
- **Balanced Metrics:** Precision (0.70) and Recall (0.67) indicate no systematic bias.

Key Weaknesses

- **Small Lesion Struggles:** Poor performance on patients with diffuse lesions (<0.40 Dice).

Future Improvements Can Address Current Limitations

Train the model with a 3d Unet

Ensure better understanding for the model

Expanded Context

Stack 5 adjacent slices instead of 3 to capture more inter-slice information.

Division for length of lesions

Divide the patient between patient with no lesion, low, medium and high volume

Conclusion

This lightweight 2.5D U-Net achieves **0.67 mean Dice** on unseen test patients, demonstrating the feasibility of automated MS lesion segmentation within constrained compute environments.